Lyapunov functions for semi-implicit dynamics with nonlinear friction

Abstract

We study a semi-implicit gradient system with pure power friction. One modified energy gives two results under distinct assumptions. Near a compact Morse–Bott minimum manifold it yields small-energy capture, finite length and convergence to one minimum. With global Hessian and Hessian-Lipschitz bounds, a structural step restriction gives coercivity and descent from every original phase, without minimum geometry or an assumed bounded orbit. This implies boundedness, summable velocity and gradient powers, and a connected critical limit set; global point convergence is not asserted. The proof adds the friction potential to a standard symplectic correction and sums two successive energies. A sign estimate controls arbitrary velocities, and quadratic and cubic force-remainder bounds give global absorption. The conservative correction and finite-step construction are inherited. A separate scalar example with mixed linear–cubic drag and quartic potential admits direct physical-energy descent for one fixed step interval and every initial phase, despite unbounded curvature. A further scalar C^4 strongly convex class with pure friction $|v|^{3/2}v$ admits arbitrary-phase convergence under an explicit structural step bound. A physical-energy cap first gives global capture; an exact cubic correction and a finite observation window then give local dissipation. A continuous strict global function is obtained from finite future entry segments. For mixed radial drag, a further physical-energy estimate treats any positive-leading quartic polynomial in finite dimension, including nonconvex potentials with unbounded negative curvature. A coefficient-only step bound yields arbitrary-phase capture, a common compact absorbing sublevel and approach to the critical set.

Keywords: discrete dynamical system; Lyapunov function; nonlinear friction; finite length; Morse–Bott minimum manifold; global asymptotic stability

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1 The system and the Lyapunov function

We first state and prove the local result in Sections 1–4. Section 5 replaces its minimum geometry and small initial energy by global curvature bounds and a further step restriction. Both results use the same state function, but their trajectory conclusions are different. Section 6 treats a separate mixed-drag scalar model by its physical energy. Section 7 gives an arbitrary-phase scalar result for pure friction with exponent 3/2 under strong convexity and global derivative bounds. Section 8 treats mixed radial drag and nonconvex positive-leading quartic potentials by a coefficient-only physical energy.

Let $F : \mathbb{R}^d \to \mathbb{R}$ be a C^3 coercive function, where $d \geq 2$, and put

$$F_* = \min_{\mathbb{R}^d} F, \qquad f = F - F_*, \qquad g = \nabla F.$$

Assume that the full minimum set

$$M = \operatorname*{argmin}_{\mathbb{R}^d} F \tag{1}$$

is a nonempty compact embedded C^2 manifold without boundary. We impose the Morse–Bott minimum condition

$$\ker \nabla^2 F(z) = T_z M, \qquad \langle \nabla^2 F(z)\xi, \xi\rangle \geq \kappa \|\xi\|^2 \quad (\xi \perp T_z M,\ z \in M) \tag{2}$$

for one constant $\kappa > 0$. The set M may be nonisolated or disconnected.

Fix a closed Euclidean ball B whose interior contains the closed unit neighborhood of M. Choose $L \geq 1$ such that

$$\|\nabla^2 F(u)\| \leq L \quad (u \in B), \qquad 0 < h, \qquad h^2 L \leq 1. \tag{3}$$

The norm in (3) is the operator norm. No global Hessian bound is assumed. The ball, its curvature bound and the step are fixed before the initial state is chosen.

For $0 < \alpha < 1$, consider the original phase iteration

$$\begin{aligned} u_{n+1} &= u_n + h v_{n+1}, \\ v_{n+1} &= v_n - h\|v_{n+1}\|^{\alpha} v_{n+1} - h g(u_n), \qquad n \geq 0. \end{aligned} \tag{4}$$

Here u_0 and v_0 are independent initial variables. This is the semi-implicit scheme in Horsin and Jendoubi's author version [1, Section 5.2, equation (54)]. It is also the scheme in Attouch, Boţ and Csetnek [2, Section 8]: their position variables are related by

$$u_n = x_{n+1}, \qquad v_n = \frac{x_{n+1} - x_n}{h}, \qquad x_0 = u_0 - h v_0, \quad x_1 = u_0.$$

Thus arbitrary phase starts in (4) correspond to arbitrary pairs of their initial positions.

Write $T = T_h$ for the phase map in (4). Define

$$\begin{aligned} E(u, v) &= f(u) + \frac{1}{2}\|v\|^2, \\ \psi(v) &= \frac{\|v\|^{\alpha+2}}{\alpha + 2}, \\ H(u, v) &= E(u, v) - \frac{h}{2}\langle g(u), v\rangle + \frac{h}{2}\psi(v), \\ \Phi(u, v) &= H(u, v) + H\big(T(u, v)\big). \end{aligned} \tag{5}$$

The last expression uses the uniquely determined next state. We will prove local coercivity and descent of Φ directly from (4).

Theorem 1. *Under* (1)–(3)*, the map T is globally well defined and bijective. There exist constants*

$$\delta, c_*, C_*, c_d, C_E > 0$$

and a fixed bounded phase neighborhood $\mathcal{U}$ of $M \times \{0\}$ such that every original state with $E(u_0, v_0) \leq \delta$ generates an orbit in $\mathcal{U}$. For all $n \geq 0$, put $\Phi_n = \Phi(u_n, v_n)$. Then

$$E(u_n, v_n) \leq C_E E(u_0, v_0), \tag{6}$$

$$c_*\big(\operatorname{dist}(u_n, M)^2 + \|v_n\|^2\big) \leq \Phi_n \leq C_*\big(\operatorname{dist}(u_n, M)^2 + \|v_n\|^2\big), \tag{7}$$

$$\Phi_n - \Phi_{n+1} \geq c_d \Phi_n^{1+\alpha/2}. \tag{8}$$

All constants are independent of the individual initial state. If $E(u_0, v_0) = 0$, the orbit is stationary. Otherwise every Φ_n is positive. With

$$r = \alpha/2, \qquad k = rc_d,$$

one has

$$\Phi_n \le [\Phi_0^{-r} + kn]^{-1/r}, \qquad \|v_n\| \le c_*^{-1/2}[\Phi_0^{-r} + kn]^{-1/\alpha}. \tag{9}$$

The position orbit has finite polygonal length and converges to one point $u_\infty \in M$. Moreover,

$$\|u_n - u_\infty\| \le \frac{hc_*^{-1/2}\alpha}{k(1-\alpha)}[\Phi_0^{-r} + kn]^{-(1-\alpha)/\alpha}, \qquad n \ge 0. \tag{10}$$

The theorem concerns a full small-energy neighborhood in the original phase space. It does not assume boundedness or a trapping time. It does not prescribe which point of M is selected. The restriction (3) is a sufficient bound, and no sharp step threshold is asserted. The physical energy E need not decrease, as the exact example in Section 9 will show.

The main distinction from [2, Theorem 9] is the damping law. That theorem assumes a positive quadratic lower bound for the damping potential. For the present ψ,

$$\frac{\psi(v)}{\|v\|^2} = \frac{\|v\|^\alpha}{\alpha + 2} \longrightarrow 0 \quad \text{as } v \longrightarrow 0.$$

Shrinking the initial neighborhood does not supply the missing positive lower bound. We instead establish a two-step decrement and the original orbit's capture.

The energy construction uses established ingredients. The conservative correction $-h\langle g(u), v\rangle/2$ appears in symplectic modified energies [5, Proposition 9]. Wibisono, Tao and Piliouras [6, Lemma 4.2] give its exact conservative Bregman-commutator identity, with the cubic estimate in their supplementary Lemma A.2. The sum in (5) is the standard finite-step Lyapunov construction; see [7, Theorem 2.3.1], which attributes it to [8]. Discrete observability and nonlinear dissipation have also been combined to derive energy decay for other schemes [9]. Our proof supplies the friction correction, the two-step estimate and the local capture argument for (4). We prove the needed geometric and scalar estimates to keep the dependence of the orbit argument explicit.

2 The phase map and local energy bounds

Put $p = 2 + \alpha$ and $D = \nabla\psi$, so that $D(v) = \|v\|^\alpha v$. Given an input (u, v), the next velocity is determined by

$$w + hD(w) = v - hg(u), \qquad x = u + hw. \tag{11}$$

The scalar function $s \mapsto s + hs^{1+\alpha}$ is strictly increasing from zero to infinity. Since the vector on the left of (11) is a positive multiple of w, this proves existence and uniqueness for every input. It also gives

$$\|w\| \le \|v - hg(u)\| \le \|v\| + h\|g(u)\|. \tag{12}$$

The resulting map is continuous. Its inverse is explicitly

$$T^{-1}(x, w) = \big(x - hw,\ w + hD(w) + hg(x - hw)\big). \tag{13}$$

In particular, every $(z, 0)$ with $z \in M$ is a fixed point whose only preimage is itself.

For a phase state $s = (u, v)$, write

$$q(s) = \big(\operatorname{dist}(u, M)^2 + \|v\|^2\big)^{1/2}.$$

The following uniform estimates are standard local consequences of the Morse–Bott condition; compare [10]. We include their proof because the local curvature bound in (3) must be sufficient throughout the argument.

Lemma 1. *There is $r_0 \in (0, 1/2)$ such that, for $\operatorname{dist}(u, M) < r_0$,*

$$\frac{\kappa}{4} \operatorname{dist}(u, M)^2 \le f(u) \le \frac{L}{2} \operatorname{dist}(u, M)^2, \tag{14}$$

$$\frac{\kappa}{2} \operatorname{dist}(u, M) \le \|g(u)\| \le L \operatorname{dist}(u, M), \qquad \|g(u)\|^2 \le 2Lf(u). \tag{15}$$

For these positions and $\|v\| \le 1$,

$$\frac{1}{2} E(u, v) \le H(u, v) \le 2E(u, v). \tag{16}$$

Moreover, if

$$m = \min\{\kappa/4, 1/2\}, \qquad M_E = L/2, \tag{17}$$

then $mq(u, v)^2 \le E(u, v) \le M_E q(u, v)^2$ in the position tube.

Proof. Choose a closest point $z \in M$ to u and put $\xi = u - z$. Compactness gives such a point, and the absence of boundary implies $\xi \perp T_z M$. Also $g(z) = 0$. Taylor expansion and (2), uniformly over $z \in M$, give

$$f(u) = \frac{1}{2}\langle \nabla^2 F(z)\xi, \xi\rangle + O(\|\xi\|^3), \qquad g(u) = \nabla^2 F(z)\xi + O(\|\xi\|^2).$$

The Hessian is symmetric and preserves the normal space. Reducing a uniform radius gives the lower bounds in (14)–(15). Integrating the Hessian on $[z, u] \subset B$ gives their upper bounds. The uniform remainders follow from the bounded third derivative on the fixed ball.

For the last inequality in (15), put $y = u - g(u)/L$. The preceding upper bound implies

$$\operatorname{dist}(y, M) \le 2 \operatorname{dist}(u, M) < 1.$$

Thus u, y and their segment are in B. The descent estimate on this segment yields

$$F_* \le F(y) \le F(u) - \frac{1}{2L}\|g(u)\|^2.$$

This argument uses the global minimum value but only the local Hessian bound.

Next, (15) and $h\sqrt{L} \le 1$ imply

$$\frac{h}{2}\big|\langle g(u), v\rangle\big| \le h\sqrt{L}\sqrt{f(u)}\frac{\|v\|}{\sqrt{2}} \le \frac{1}{2}E(u,v).$$

The damping-potential term is nonnegative. If $\|v\| \le 1$, then $h \le 1$ and $p > 2$ give

$$\frac{h}{2}\psi(v) = \frac{h}{2p}\|v\|^p \le \frac{h}{p}E(u,v) \le \frac{1}{2}E(u,v).$$

This proves (16). The final comparison follows from (14) and $L \ge 1$. ∎

3 The decrement over two steps

For a differentiable convex function ψ, denote its Bregman divergence by

$$B_\psi(a,b) = \psi(a) - \psi(b) - \langle D(b), a - b\rangle.$$

The reflected argument in the next identity is important.

Lemma 2. *For every actual step $(u,v) \mapsto (x,w)$ in (11),*

$$H(x,w) - H(u,v) = R(u,w) - \frac{h}{2}B_\psi(v,-w), \tag{18}$$

where

$$R(u,w) = F(u+hw) - F(u) - \frac{h}{2}\langle g(u) + g(u+hw), w\rangle. \tag{19}$$

If the position segment is in B and $K_3 = \sup_B\|\nabla^3 F\|$, then

$$|R(u,w)| \le \frac{K_3 h^3}{12}\|w\|^3. \tag{20}$$

Proof. Omitting the damping-potential term from H, its change along the step is

$$R(u,w) - \frac{h}{2}\langle D(w), v + w\rangle.$$

Indeed, substitute $v = w + hD(w) + hg(u)$ into the kinetic and cross terms and use $x - u = hw$. Adding $h[\psi(w) - \psi(v)]/2$ gives (18), because ψ is even and D is odd. In particular,

$$B_\psi(v,-w) = \psi(v) - \psi(w) + \langle D(w), v + w\rangle.$$

For the remainder, integration by parts in the scalar trapezoid formula gives

$$R(u,w) = -\frac{h^3}{2}\int_0^1 t(1-t)\nabla^3 F(u+thw)[w,w,w]\,dt.$$

Since $\int_0^1 t(1-t)/2\,dt = 1/12$, this proves (20). Its sign is not fixed. The conservative identity and this remainder estimate agree with the commutator formulas in [6, Lemma 4.2 and supplementary Lemma A.2]; the additional friction term in (18) follows from the actual damped update. ∎

Lemma 3. *For all $a, b \in \mathbb{R}^d$,*

$$B_\psi(a,b) \geq \beta\|a-b\|^p, \qquad \beta = 8^{-1-\alpha} > 0. \tag{21}$$

Proof. The function ψ is C^2, including at zero, and $\nabla^2\psi(z) \succeq \|z\|^\alpha \,\mathrm{Id}$. If $a \neq b$, Taylor's integral formula and projection onto $(a-b)/\|a-b\|$ give

$$B_\psi(a,b) \geq \|a-b\|^p \int_0^1 (1-t)|A+t|^\alpha\,dt$$

for a real number A. On $[0,1/2]$ the weight is at least $1/2$. The set where $|A+t| < 1/8$ has length at most $1/4$. The remaining set has length at least $1/4$, so the integral is at least $8^{-1-\alpha}$. The case $a = b$ is immediate. ∎

The divergence in (18) controls $v+w$, and it vanishes when $v = -w$. Two successive steps control this alternating direction.

Lemma 4. *There exist $\eta \in (0,1]$ and $d_0 > 0$ such that the following holds. Suppose three successive states*

$$(u,a), \qquad (x,b) = T(u,a), \qquad (y,c) = T^2(u,a)$$

have positions in the r_0-tube of Lemma 1 and velocities of norm at most η. Put $s_0 = a+b$, $s_1 = b+c$ and $S = \|s_0\| + \|s_1\|$. Then

$$q(u,a) \leq JS, \qquad J = 2 + \frac{8}{\kappa h}, \tag{22}$$

and

$$H(u,a) - H(y,c) \geq d_0 S^p. \tag{23}$$

One may take $d_0 = h\beta 2^{-p-1}$ after choosing η sufficiently small.

Proof. All position segments lie in the convex ball B. Define

$$A = \int_0^1 \nabla^2 F(u+thb)\,dt, \qquad g(x) - g(u) = hAb, \qquad \|A\| \leq L.$$

Subtract the two original velocity equations. The resulting exact identity is

$$s_0 + s_1 = (4\,\mathrm{Id} - h^2 A)b + h[D(b) - D(c)]. \tag{24}$$

Since $c = s_1 - b$ and D is odd, it can also be written as

$$(4\,\mathrm{Id} - h^2 A)b + 2hD(b) = s_0 + s_1 + h[D(c) - D(-b)].$$

The matrix A is symmetric and $4\,\mathrm{Id} - h^2 A \succeq 3\,\mathrm{Id}$. On the unit velocity ball, D is $(1+\alpha)$-Lipschitz. Taking the scalar product with b therefore gives

$$3\|b\|^2 + 2h\|b\|^p \le \|b\|\big(\|s_0\| + [1 + h(1+\alpha)]\|s_1\|\big).$$

As $h \le 1$ and $\alpha < 1$, this implies, including when $b = 0$,

$$\|b\| \le S, \qquad \|a\| \le 2S, \qquad \|c\| \le 2S. \tag{25}$$

The first original velocity equation gives

$$\|g(u)\| \le \frac{\|a\| + \|b\|}{h} + \|D(b)\| \le \frac{4}{h} S,$$

where $\|b\| \le \eta \le 1$ was used. The gradient lower bound in (15) now proves (22).

Adding (18) over two steps and using Lemma 3, we obtain

$$\begin{aligned} H(u,a) - H(y,c) &\ge \frac{h\beta}{2}\big(\|s_0\|^p + \|s_1\|^p\big) - \frac{K_3 h^3}{12}\big(\|b\|^3 + \|c\|^3\big) \\ &\ge h\beta 2^{-p} S^p - \frac{3K_3 h^3}{4} S^3. \end{aligned} \tag{26}$$

Here (25) bounds the sum of cubes by $9S^3$. Also $S \le 4\eta$. Choose $\eta > 0$ so small that

$$\frac{3K_3 h^2}{4}(4\eta)^{1-\alpha} \le \beta 2^{-p-1}. \tag{27}$$

If $K_3 = 0$, this condition is automatic. Otherwise its positive exponent $1 - \alpha$ makes such a choice possible. Substitution in (26) proves (23). ∎

4 Capture of the original orbit and its limit

We now prove that the local estimates apply at every original index. The argument uses a fixed neighborhood of phase states whose first two successors are already defined by the global map.

Proof of Theorem 1. Global solvability and bijectivity follow from (11)–(13). Fix r_0 and η from the preceding lemmas and set

$$\begin{aligned} V &= \{(u,v) : \operatorname{dist}(u, M) < r_0,\ \|v\| < \eta\}, \\ \Omega &= V \cap T^{-1}(V) \cap T^{-2}(V). \end{aligned} \tag{28}$$

Continuity of T and its fixed points $(z, 0)$ show that Ω is a bounded open neighborhood of the entire compact set $M \times \{0\}$. This definition uses only two actual evaluations of the global map. It assumes nothing about an infinite orbit.

For an input in the position tube, (12) and (15) imply

$$q(Ts) \le K q(s), \qquad K = 1 + (h+1)(1 + hL). \tag{29}$$

Indeed, $\|w\| \le (1+hL)q(s)$ and $\operatorname{dist}(x,M) \le \operatorname{dist}(u,M)+h\|w\|$; bounding the Euclidean phase norm by the sum gives (29). On Ω, both s and Ts are in V, so Lemma 1 yields

$$c_* q(s)^2 \le \Phi(s) \le C_* q(s)^2 \le C_\Phi E(s), \tag{30}$$

where we may take

$$c_* = m/2, \qquad C_* = 2M_E(1+K^2), \qquad C_\Phi = C_*/m. \tag{31}$$

Since Ts and $T^2 s$ are also in V, we have the separate bound

$$\Phi(Ts) = H(Ts) + H(T^2 s) \ge \frac{1}{2} E(Ts) \quad (s \in \Omega). \tag{32}$$

This inequality does not require $Ts \in \Omega$.

For each individual $s \in \Omega$, the first three states satisfy Lemma 4. The finite-step sum in (5) therefore gives

$$\begin{aligned} \Phi(s) - \Phi(Ts) &= H(s) - H(T^2 s) \\ &\ge d_0 J^{-p} q(s)^p \ge c_d \Phi(s)^{1+\alpha/2}, \end{aligned} \tag{33}$$

with $c_d = d_0 J^{-p} C_*^{-p/2} > 0$. Thus the estimate is established on a fixed phase neighborhood before it is iterated.

The physical energy E is continuous, nonnegative and coercive on $\mathbb{R}^{2d}$, and its zero set is exactly $M \times \{0\}$. Consequently there is $\epsilon > 0$ such that

$$\{s : E(s) \le \epsilon\} \subset \Omega. \tag{34}$$

For otherwise a sequence outside Ω with energy tending to zero would have a convergent subsequence in a compact energy sublevel. Its limit would be in $M \times \{0\} \subset \Omega$, a contradiction. This is where the full-set condition (1) is used.

Choose

$$\delta = \frac{\epsilon}{4\max\{1, C_\Phi\}}, \qquad C_E = 2C_\Phi. \tag{35}$$

Let $s_0 = (u_0, v_0)$ be any original pair with $E(s_0) \le \delta$. Then $s_0 \in \Omega$. Suppose inductively that $s_n \in \Omega$ and $\Phi(s_n) \le \Phi(s_0)$. Applying (33) and then (32) gives

$$E(s_{n+1}) \le 2\Phi(s_{n+1}) \le 2\Phi(s_n) \le 2\Phi(s_0) \le 2C_\Phi E(s_0) \le \epsilon/2.$$

Equation (34) now puts the actual successor in Ω, closing the induction. Taking $\mathcal{U} = \Omega$ proves capture, (6), (7) and (8). In particular, boundedness has been derived from the original initial-energy condition.

If $E(s_0) = 0$, then $s_0 \in M \times \{0\}$ and the orbit is stationary. For a positive initial energy, bijectivity and (13) show that no state can enter $M \times \{0\}$ at a finite index. Hence every Φ_n is positive by (7). The convexity of $t \mapsto t^{-r}$, with $r = \alpha/2$, and (8) give

$$\Phi_{n+1}^{-r} - \Phi_n^{-r} \ge r\Phi_n^{-r-1}(\Phi_n - \Phi_{n+1}) \ge r c_d = k.$$

Summing from $n = 0$ proves the first bound in (9). The second follows from $c_*\|v_n\|^2 \le \Phi_n$.

Finally, $1/\alpha > 1$. For every $n \ge 0$, the original position update and the decreasing scalar summand yield

$$\begin{aligned}\sum_{j=n}^{\infty}\|u_{j+1}-u_j\| &\le hc_*^{-1/2}\sum_{j=n}^{\infty}[\Phi_0^{-r}+k(j+1)]^{-1/\alpha}\\ &\le hc_*^{-1/2}\int_n^{\infty}[\Phi_0^{-r}+kt]^{-1/\alpha}\,dt\\ &= \frac{hc_*^{-1/2}\alpha}{k(1-\alpha)}[\Phi_0^{-r}+kn]^{-(1-\alpha)/\alpha}.\end{aligned}$$

Thus the position orbit is Cauchy and has finite polygonal length. Equations (7) and (9) show that the velocity and the distance to M tend to zero. Closedness of M gives $u_\infty \in M$, and the preceding length estimate proves (10). ∎

5 Global control from arbitrary original states

In this section we replace the minimum-manifold and local-curvature assumptions of Theorem 1 by a separate global class. Let $d \ge 1$ be finite, and let $F : \mathbb{R}^d \to \mathbb{R}$ be coercive and C^3. Assume that known constants $L \ge 1$ and $K \ge 0$ satisfy

$$\|\nabla^2 F(u)\| \le L, \qquad \|\nabla^2 F(u) - \nabla^2 F(x)\| \le K\|u-x\| \quad (u, x \in \mathbb{R}^d). \tag{36}$$

No convexity, quadratic growth, isolated-critical-point, KL or Morse–Bott hypothesis is imposed. The notation $f = F - \min F$, $g = \nabla F$, $p = 2+\alpha$ and $D(v) = \|v\|^\alpha v$ is retained, with $0 < \alpha < 1$. Consider the same original map (4) and the same functions E, H, Φ in (5). Fix $h > 0$ such that

$$h^2 L \le 1, \qquad 5 \cdot 16^{1+\alpha} h^{1+\alpha} K^\alpha L^{1-\alpha} \le 1. \tag{37}$$

If $K = 0$, the second left-hand side is zero. These conditions admit positive steps and depend only on α, L, K, independently of the original phase state. The normalization by $\min F$ does not enter the iteration or the step choice.

Theorem 2. *Under (36)–(37), T is a global homeomorphism and H, Φ are continuous, nonnegative and coercive on $\mathbb{R}^d \times \mathbb{R}^d$. For every phase $s = (u, v)$,*

$$H(s) \ge \frac{1}{2}E(s), \qquad \Phi(s) \ge \frac{E(s)+E(Ts)}{2} \ge \frac{1}{2}E(s). \tag{38}$$

For an arbitrary original (u_0, v_0), define

$$S_n = \|v_n + v_{n+1}\| + \|v_{n+1} + v_{n+2}\|, \qquad c_\alpha = 2^{-6-4\alpha}. \tag{39}$$

Then, at every $n \ge 0$,

$$\Phi(u_n, v_n) - \Phi(u_{n+1}, v_{n+1}) \ge hc_\alpha S_n^p. \tag{40}$$

Successive energy values are equal exactly at fixed phases $(z,0)$ with $g(z)=0$.

Put $P_0=\Phi(u_0,v_0)$ and $A_0=3/h+(2\sqrt{P_0})^\alpha$. The full original phase orbit is bounded, and

$$0\le \Phi(u_n,v_n)\le P_0, \qquad E(u_n,v_n)\le 2P_0, \tag{41}$$

$$\sum_{n=0}^{\infty} S_n^p \le \frac{P_0}{hc_\alpha}, \qquad \sum_{n=0}^{\infty}\|v_{n+1}\|^p \le \frac{P_0}{hc_\alpha}, \tag{42}$$

$$\sum_{n=0}^{\infty}\|g(u_n)\|^p \le \frac{A_0^p P_0}{hc_\alpha}. \tag{43}$$

In particular, $v_n\to 0$ and $g(u_n)\to 0$. If $P_0=0$, the orbit is stationary at a global minimum.

If $\ell=\lim_n \Phi(u_n,v_n)$, then

$$F(u_n)\longrightarrow \min F+\ell/2. \tag{44}$$

The phase omega-limit set is nonempty, compact and connected, and is contained in

$$\{(z,0): g(z)=0,\ F(z)=\min F+\ell/2\}. \tag{45}$$

No point-convergence, finite-length, rate or minimum-selection conclusion is asserted under these global hypotheses alone.

The global result derives compactness before using any critical-set information. Its sublevel bound depends on the original phase through P_0; it is not an absorbing ball common to all starts. It complements Theorem 1: it adds global curvature regularity while removing small initial energy and minimum geometry, and it has a weaker trajectory conclusion.

The formula for H, its reflected Bregman identity, and the finite-step sum are unchanged. The new estimates below control the alternating direction at arbitrary velocity amplitude and combine two global bounds on the signed force remainder. Thus replacing the local constants by global ones is not the argument.

Lemma 5. *Let (u,a), $(x,b)=T(u,a)$ and $(y,c)=T^2(u,a)$ be any three successive phases under (36) and $h^2L\le 1$. Put*

$$s_0=a+b, \qquad s_1=b+c, \qquad S=\|s_0\|+\|s_1\|.$$

Then

$$\|b\|\le S, \qquad \|a\|\le 2S, \qquad \|c\|\le 2S. \tag{46}$$

Proof. Set $A=\int_0^1 \nabla^2 F(u+thb)\,dt$. The two original velocity equations give the exact relation (24), now globally:

$$s_0+s_1=(4\,\mathrm{Id}-h^2A)b+h[D(b)-D(c)], \qquad 4\,\mathrm{Id}-h^2A\succeq 3\,\mathrm{Id}\,. \tag{47}$$

If $\|b\| \le \|s_1\|$, the first bound is immediate. Otherwise $b \neq 0$, $c = s_1 - b$, and

$$\langle D(c), b\rangle = \|c\|^\alpha(\langle s_1, b\rangle - \|b\|^2) \le 0.$$

Taking the scalar product of (47) with b gives

$$3\|b\|^2 + h\|b\|^p \le \langle s_0 + s_1, b\rangle \le S\|b\|.$$

In this case $\|b\| \le S/3$. The first bound in (46) follows in both cases, including $b = 0$. The other two follow from $a = s_0 - b$ and $c = s_1 - b$. No upper velocity bound is used. ∎

Lemma 6. *For every $u, w \in \mathbb{R}^d$, the force remainder in* (19) *satisfies*

$$|R(u,w)| \le \min\left\{\frac{Lh^2}{4}\|w\|^2, \frac{Kh^3}{12}\|w\|^3\right\}. \tag{48}$$

For the successive states in Lemma 5,

$$|R(u,b)| + |R(x,c)| \le \frac{5}{4}K^\alpha L^{1-\alpha}h^{2+\alpha}S^p \le hc_\alpha S^p. \tag{49}$$

Proof. The scalar trapezoid identity gives

$$R(u,w) = h^2\int_0^1 (1/2 - t)\langle \nabla^2 F(u + thw)w, w\rangle\, dt.$$

The integral of $|1/2 - t|$ is $1/4$, proving the quadratic bound. Since $\int_0^1 (1/2 - t)\, dt = 0$, subtracting the constant Hessian at $u + hw/2$ and using (36) gives the cubic bound: $\int_0^1 (t - 1/2)^2\, dt = 1/12$. This agrees with the conservative commutator estimate already used in Lemma 2 and attributed to [6].

By (46),

$$|R(u,b)| + |R(x,c)| \le \min\left\{\frac{5Lh^2}{4}S^2, \frac{3Kh^3}{4}S^3\right\}.$$

For nonnegative X, Y, $\min(X,Y) \le X^{1-\alpha}Y^\alpha$. Therefore the preceding expression is at most

$$\frac{5^{1-\alpha}3^\alpha}{4}K^\alpha L^{1-\alpha}h^{2+\alpha}S^{2+\alpha} \le \frac{5}{4}K^\alpha L^{1-\alpha}h^{2+\alpha}S^p.$$

The second step condition (37) gives its final bound in (49), since

$$\frac{5/4}{5\cdot 16^{1+\alpha}} = 2^{-6-4\alpha} = c_\alpha.$$

When $K = 0$, the cubic bound says $R = 0$ directly. When $S = 0$, all three velocities vanish by (46). These cases require no division by K or S. The two remainder estimates handle small and arbitrarily large velocities in one global bound. ∎

Proof of Theorem 2. The global map and inverse follow directly from (11)–(13); neither argument uses the earlier geometric hypotheses. We prove coercivity first. Applying the global descent estimate at $u - g(u)/L$ gives

$$\|g(u)\|^2 \le 2Lf(u).$$

It follows from $h\sqrt{L} \le 1$ that

$$\frac{h}{2}\big|\langle g(u), v\rangle\big| \le h\sqrt{L}\sqrt{f(u)}\frac{\|v\|}{\sqrt{2}} \le \frac{1}{2}E(u,v).$$

Since $\psi \ge 0$, this proves (38). The function E is coercive on phase space by coercivity of F and its kinetic term. Thus H and Φ are continuous and coercive. No orbit bound has yet been used.

The exact identity (18) applies to both actual steps from (u,a) to (x,b) to (y,c). The two intermediate H terms cancel, yielding

$$\begin{aligned}\Phi(u,a) - \Phi(x,b) = \frac{h}{2}\big[B_\psi(a,-b) + B_\psi(b,-c)\big] \\ - R(u,b) - R(x,c).\end{aligned} \tag{50}$$

The global power-divergence estimate (21) implies

$$\frac{h}{2}\big[B_\psi(a,-b) + B_\psi(b,-c)\big] \ge h\beta 2^{-p}S^p = 2hc_\alpha S^p.$$

Combining this with Lemma 6 proves (40). If $S = 0$, Lemma 5 gives $a = b = c = 0$, and the original recurrence gives $g(u) = 0$. Conversely such a phase is fixed. Therefore Φ decreases strictly at every nonfixed phase.

Now start at any original (u_0, v_0). Nonnegativity and the decrement give (41). The full orbit lies in the compact sublevel $E \le 2P_0$, so boundedness is a consequence. Telescoping and (46) give (42). Also $\|v_n\| \le 2\sqrt{P_0}$, and the actual old-position force satisfies

$$\begin{aligned}\|g(u_n)\| &\le \frac{1}{h}\|v_n - v_{n+1}\| + \|v_{n+1}\|^{1+\alpha} \\ &\le \left(\frac{3}{h} + (2\sqrt{P_0})^\alpha\right) S_n.\end{aligned}$$

This proves (43). In particular, both velocity and gradient tend to zero. If $P_0 = 0$, coercivity comparison (38) forces $f(u_0) = 0$ and $v_0 = 0$; the gradient vanishes at this minimum and the orbit is stationary. No division by P_0 is made.

The decreasing sequence Φ_n has a finite limit ℓ. The velocity and gradient limits imply $H(u_n, v_n) = f(u_n) + o(1)$. Moreover,

$$|f(u_{n+1}) - f(u_n)| \le h\|g(u_n)\|\|v_{n+1}\| + \frac{Lh^2}{2}\|v_{n+1}\|^2 \longrightarrow 0.$$

Hence $\Phi_n = 2f(u_n) + o(1)$, giving (44). Every phase cluster point then belongs to (45).

For completeness, write Ω for the phase omega-limit set. It is nonempty and compact by boundedness. The phase increments tend to zero, since $u_{n+1} - u_n = hv_{n+1}$ and both velocities tend to zero. Also the distance of the phase sequence to Ω tends to zero; otherwise a bounded subsequence separated from Ω would have a further limit in Ω. If Ω were disconnected, partition it into two nonempty disjoint compact sets at distance

$\delta > 0$. Eventually every phase lies in the union of their $\delta/4$-neighborhoods, and both neighborhoods are visited arbitrarily late. But successive phase increments eventually have length less than $\delta/2$, so the sequence cannot pass from one neighborhood to the other. This contradiction proves connectedness. It does not imply that Ω is a singleton. ∎

Theorem 2 establishes global energy control and compactness for (4), not the full point-convergence question after equation (54) of [1]. The inherited state function gains a global decrement from the all-amplitude velocity estimate and the two force-remainder bounds.

6 A quartic potential with mixed polynomial drag

The global curvature assumption in (36) excludes quartic potentials. A separate scalar model admits a direct physical-energy estimate. Here we replace the pure power friction by $D(v) = v + v^3$ and take

$$\begin{aligned} (1+h)v_{n+1} + hv_{n+1}^3 &= v_n - h(x_n + x_n^3), \\ x_{n+1} &= x_n + hv_{n+1}, \qquad n \geq 0. \end{aligned} \tag{51}$$

The two coordinates of $(x_0, v_0) \in \mathbb{R}^2$ are independent. Set

$$F_q(x) = \frac{x^2}{2} + \frac{x^4}{4}, \qquad E_q(x,v) = F_q(x) + \frac{v^2}{2}. \tag{52}$$

The following statement concerns only this fixed model.

Proposition 6.1. For every fixed $0 < h \leq 1/8$, the update (51) defines a continuous map on $\mathbb{R}^2$. The function E_q is positive definite and coercive, with

$$E_q(x,v) \geq \frac{x^2 + v^2}{2}. \tag{53}$$

If w is the actual next velocity at (x, v), then

$$E_q(x + hw, w) \leq E_q(x,v) - \frac{h}{8}(w^2 + w^4). \tag{54}$$

Consequently every original phase orbit is bounded and satisfies

$$\sum_{n=0}^{\infty} \bigl(v_{n+1}^2 + v_{n+1}^4\bigr) \leq \frac{8E_q(x_0, v_0)}{h}. \tag{55}$$

The full phase converges to $(0,0)$, which is globally asymptotically stable. The step interval is independent of the initial energy.

Proof. For fixed $h > 0$, the function $w \mapsto (1+h)w + hw^3$ is strictly increasing onto $\mathbb{R}$ and has a continuous inverse. Thus the actual map is globally defined and continuous. Coercivity and positive definiteness follow directly from (52).

We prove (54) before considering any orbit. Put

$$A = x + x^3 + w + w^3, \qquad B = x^2 - xw + w^2, \qquad P = \frac{A^2}{4} + 4w^4 - x^2w^2.$$

The factorization $A = (x + w)(1 + B)$ gives the exact identity

$$\begin{aligned} P = \frac{1}{4}(x+w)^2(B-1)^2 + \left(x^2 + \frac{xw}{2} - w^2\right)^2 \\ + \frac{1}{4}w^2(x+4w)^2 + \frac{1}{2}x^2w^2 \geq 0. \end{aligned} \tag{56}$$

There is no bound on x or w in this identity. The original velocity equation is $v = w + hA$. Expanding the quartic potential therefore yields

$$\begin{aligned} E_q(x + hw, w) - E_q(x, v) = & -h(w^2 + w^4) \\ & + \frac{h^2}{2}\left[(1 + 3x^2)w^2 + 2hxw^3 + \frac{h^2}{2}w^4 - A^2\right]. \end{aligned} \tag{57}$$

Write $C_h = 12 + 5h + h^2/2$. The bracket on the right equals

$$w^2 + C_h w^4 - \frac{1-h}{4}A^2 - (3+h)P - hw^2(x-w)^2.$$

Indeed this follows by substituting P and using $2xw^3 = x^2w^2 + w^4 - w^2(x-w)^2$. Consequently

$$\begin{aligned} E_q(x, v) - E_q(x + hw, w) = h\left(1 - \frac{h}{2}\right)w^2 + h\left(1 - \frac{hC_h}{2}\right)w^4 \\ + \frac{h^2(1-h)}{8}A^2 + \frac{h^2(3+h)}{2}P + \frac{h^3}{2}w^2(x-w)^2. \end{aligned} \tag{58}$$

For $0 < h \leq 1/8$, one has $C_h \leq 1617/128 < 13$ and

$$1 - \frac{h}{2} \geq \frac{15}{16}, \qquad 1 - \frac{hC_h}{2} \geq \frac{3}{16} > \frac{1}{8}.$$

All other terms in (58) are nonnegative. This proves the global estimate (54), including the upper endpoint. If $w = 0$, then $v = h(x + x^3)$ and the exact energy drop is $h^2(x + x^3)^2/2$. Thus the energy strictly decreases at every nonzero phase, including this case.

Applying the pointwise estimate to the actual orbit and telescoping gives

$$0 \leq E_q(x_N, v_N) + \frac{h}{8}\sum_{n=0}^{N-1}(v_{n+1}^2 + v_{n+1}^4) \leq E_q(x_0, v_0), \qquad N \geq 1.$$

Coercivity now gives boundedness of the original phase. Letting N tend to infinity proves (55) and $v_n \to 0$. The unchanged update then gives

$$x_n + x_n^3 = \frac{v_n - v_{n+1}}{h} - v_{n+1} - v_{n+1}^3 \longrightarrow 0.$$

Since $|x + x^3| \geq |x|$, the whole position sequence tends to zero.

Finally, if $r^2 = x^2 + v^2$ and $r < 1$, then $E_q(x, v) \leq 3r^2/4$. For any $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/2\}$. An initial phase of norm less than δ has every future norm squared bounded by $2E_q(x_0, v_0) < 3\delta^2/2 < \varepsilon^2$. This proves Lyapunov stability. Together with global attraction it gives the claimed asymptotic stability. ∎

The mixed drag in (51) already belongs to the general algorithm in [2, Section 8]. Its Theorem 9 assumes bounded iterates and a step restriction involving the gradient Lipschitz constant on their attained ball. Remark 6(i) derives boundedness from coercivity and a globally Lipschitz gradient. Here $F_q''(x) = 1 + 3x^2$ is unbounded; Proposition 6.1 instead derives capture from a global identity with one initial-data-independent step interval. Polarized Duffing discretizations also have arbitrary-start convergence results [4, Proposition 3], but use a different force and phase history. We assert neither a new integrator nor an exhaustive priority or maximal-step result. The mixed-drag proposition does not extend the preceding pure-power theorems to unbounded curvature.

7 Global convergence for scalar degenerate friction

We now return to pure power friction, with a different exponent and a stronger scalar potential hypothesis. Throughout this section, let $F \in C^4(\mathbb{R})$ satisfy

$$0 < \mu \leq F''(u) \leq L, \qquad |F'''(u)| \leq K_3, \qquad |F''''(u)| \leq K_4 \quad (u \in \mathbb{R}), \tag{59}$$

where $K_3, K_4 \geq 0$ are finite. Translate the unique minimizer to zero and subtract its value, so that $F(0) = F'(0) = 0$. Write $g = F'$, $z = (u, v)$ and

$$w + h|w|^{3/2}w = v - hg(u), \qquad x = u + hw, \qquad T_h(u, v) = (x, w). \tag{60}$$

Thus the exponent in (4) is now $\alpha = 3/2$. The two original phase coordinates remain independent. Put

$$p = \frac{7}{2}, \qquad E(u, v) = F(u) + \frac{1}{2}v^2, \qquad \psi(v) = \frac{|v|^p}{p}.$$

Theorem 3. *Under* (59)*, define the positive structural constant* h_* *by* (66) *below. For every fixed* $0 < h \leq h_*$*, the map* (60) *has a unique successor at every phase. Every original orbit* $z_n = T_h^n z_0$*,* $z_0 \in \mathbb{R}^2$*, satisfies*

$$z_n \longrightarrow (0, 0), \qquad \sum_{n=0}^{\infty} \|z_n\|^{7/2} < \infty. \tag{61}$$

The origin is globally asymptotically stable. The physical-energy cap in (72) *is globally coercive and nonincreasing. The function* $\mathcal{V}_h$ *defined in* (88) *is continuous, positive definite and globally coercive, with*

$$\mathcal{V}_h(T_h z) - \mathcal{V}_h(z) = -\ell_h(z) < 0 \quad (z \neq 0). \tag{62}$$

The step restriction depends only on μ, L, K_3, K_4, and not on the initial phase or its energy.

The argument first proves physical capture, then a local finite-window decrement for the actual map. The global strict function uses a state-dependent finite future segment, as specified below; it is not a finite-history or polynomial formula. Finite-step summation and local-to-global Lyapunov constructions are established tools [7, 8, 12]. General converse theory also supplies continuous coercive strict functions once global asymptotic stability is known [13, Theorem 5]. We prove the needed capture, window estimate and continuity directly. In particular, no converse theorem is used to assume the missing trajectory conclusion.

7.1 Constants and phase functions

The following conservative constants make the step interval explicit. They are defined in the displayed order:

$$\begin{aligned}
&m_0 = \min\{\mu, 1\}, && M_0 = \max\{L, 1\}, && B_0 = \sqrt{3M_0/m_0},\\
&\tau = 4/\sqrt{\mu}, && T_0 = \tau + 2, && K_A = 2/m_0,\\
&c_0 = 3\tau/(4K_A), && \varepsilon_0 = \min\{1, \sqrt{c_0/T_0}/4\}, &&\\
&c_w = 1 + L, && c_z = 1 + 2c_w, && C_P = K_3(5L+3)/24,\\
&Q_0 = K_3/2 + c_w^{5/2}, && C_N = 2Q_0, &&
\end{aligned} \tag{63}$$

$$\begin{aligned}
C_R &= 1 + \frac{K_4 c_w^3(1+c_w)}{12} + \frac{K_3 c_w^2 Q_0}{4} + 10 C_P c_z^2 Q_0,\\
\rho &= \min\left\{\frac{1}{4B_0}, \frac{\varepsilon_0}{8B_0^3 T_0 C_N}, \frac{m_0}{32 B_0(1+C_P)}\right\},\\
d_0 &= \frac{2^{2-p}}{2p} T_0^{1-p/2}(c_0/4)^{p/2}, \qquad b_2 = \frac{m_0\rho^2}{2}, \qquad b_1 = \frac{b_2}{2}.
\end{aligned} \tag{64}$$

For the outer capture bound, set

$$\begin{aligned}
&W_0 = (L/2)^{2/3}, && V_0 = (1+\sqrt{L})W_0,\\
&G_0 = \sqrt{L}\,W_0 + W_0^{5/2}, && U_0 = G_0/\mu,\\
&C_B = \max\left\{\frac{L}{2}U_0^2 + \frac{1}{2}V_0^2, \frac{L}{2}(U_0+W_0)^2 + \frac{1}{2}W_0^2\right\}. &&
\end{aligned} \tag{65}$$

Finally, define

$$h_* = \min\left\{1, \frac{1}{\sqrt{L}}, \sqrt{\frac{d_0}{2C_R T_0 (2B_0)^4 \sqrt{\rho}}}, \left(\frac{b_1}{2C_B}\right)^{3/4}\right\}. \tag{66}$$

All denominators are positive, including when $K_3 = 0$ or $K_4 = 0$. Thus $h_* > 0$. Fix h in this interval and put

$$B_h = C_B h^{4/3}, \qquad N_h = \lceil \tau/h \rceil, \qquad a = F''(0), \quad b = F'''(0). \tag{67}$$

In particular,

$$h \le 1, \quad ah^2 \le Lh^2 \le 1, \quad B_h \le b_1/2, \quad \tau \le N_h h, \quad (N_h + 1)h \le T_0. \tag{68}$$

The local corrected energy and its finite future sum are

$$\begin{aligned} H_h(u,v) &= E(u,v) - \frac{h}{2} g(u)v + \frac{h}{2}\psi(v), \\ P_h(u,v) &= \frac{bh^2 u(2au^2 - 3ahuv + 3v^2)}{12(3 - ah^2)}, \qquad J_h = H_h + P_h, \end{aligned} \tag{69}$$

$$\mathcal{W}_h(z) = \sum_{j=0}^{N_h - 1} J_h(T_h^j z). \tag{70}$$

Here $3 - ah^2 \ge 2$. Neither J_h nor $\mathcal{W}_h$ is asserted to be globally positive or coercive.

7.2 Physical capture before the local argument

The scalar function $w \mapsto w + h|w|^{3/2}w$ is continuous, strictly increasing and onto. Its derivative is $1 + (5h/2)|w|^{3/2} \ge 1$. Hence (60) defines a unique continuous map on $\mathbb{R}^2$, with $T_h(0) = 0$. The inverse friction solve fixes zero and is nonexpansive. Strong convexity also gives

$$\frac{\mu}{2}u^2 \le F(u) \le \frac{L}{2}u^2, \quad \mu|u| \le |g(u)| \le L|u|, \quad \frac{m_0}{2}\|z\|^2 \le E(z) \le \frac{M_0}{2}\|z\|^2. \tag{71}$$

Lemma 7. *For every phase,*

$$E(T_h z) \le \max\{E(z), B_h\}, \qquad E(z) > B_h \implies E(T_h z) < E(z). \tag{72}$$

Every orbit is bounded and enters $\{E \le b_1\}$ in finitely many steps. The sets $\{E \le b_1\}$ and $\{E \le b_2\}$ are invariant, and entrance times into the former have a common finite bound on each compact set of initial phases.

Proof. For $T_h(u,v) = (x,w)$, Taylor's upper curvature bound and the exact velocity equation yield

$$E(x,w) - E(u,v) \le -h|w|^{7/2} - \frac{1}{2}(w - v)^2 + \frac{Lh^2}{2}w^2. \tag{73}$$

Suppose this actual energy increment is nonnegative. If $w = 0$, the inequality forces $v = 0$, and the step then gives $u = 0$. Otherwise,

$$|w| \le W_0 h^{2/3}, \qquad |w - v| \le \sqrt{L}\, h|w|.$$

The implicit equation and (71) therefore imply

$$\begin{aligned} |v| &\le V_0 h^{2/3}, \\ |g(u)| &\le \sqrt{L}|w| + |w|^{5/2} \le G_0 h^{2/3}, \\ |u| &\le U_0 h^{2/3}, \qquad |x| \le (U_0 + W_0) h^{2/3}. \end{aligned}$$

We used $h \le 1$ to bound $h^{5/3}$ by $h^{2/3}$. Both endpoint energies are thus at most B_h, by (65). If the actual increment is negative, its next energy is already smaller. These two alternatives prove (72), including exclusion of equality above the band.

Consequently $\max\{E, B_h\}$ is continuous, coercive and nonincreasing. This proves boundedness before any local argument. Since $B_h < b_1 < b_2$, both stated sublevels are invariant. For a compact initial set, choose $R \ge b_2$ above all its initial energies. All future energies remain at most R. On the compact annulus $b_1 \le E \le R$, the continuous function $E - E \circ T_h$ is strictly positive by (72); let its minimum be $\delta_R > 0$. Until entrance, every step loses at least δ_R. Nonnegativity of E gives a common finite entrance bound, for example $\lfloor R/\delta_R \rfloor + 1$. This proves the lemma without an assumed bounded trajectory. ∎

7.3 A local window of actual dissipation

For readability suppress the subscript h on T, J, H, P in this subsection. The exact linear comparison map is

$$A(u, v) = ((1 - ah^2)u + hv,\ v - hau).$$

It preserves the quadratic form

$$Q(u, v) = \frac{1}{2}(au^2 + v^2 - hauv), \qquad \frac{m_0}{4}\|z\|^2 \le Q(z) \le \frac{3M_0}{4}\|z\|^2. \tag{74}$$

Thus $\|A^j\| \le B_0$ for all integers $j \ge 0$.

Lemma 8. *For $\|z\| \le \rho$ and $0 \le j \le N_h + 1$,*

$$\|T^j z\| \le 2B_0\|z\|. \tag{75}$$

Moreover,

$$J(T^{N_h} z) - J(z) \le -\frac{d_0}{2}\|z\|^{7/2}, \tag{76}$$

$$\mathcal{W}_h(z) \ge \frac{m_0}{8}\|z\|^2, \qquad \mathcal{W}_h(z) - \mathcal{W}_h(Tz) \ge \frac{d_0}{2}\|z\|^{7/2}. \tag{77}$$

Proof. We first compare the actual orbit with $A^j z$, retaining a whole physical-time window. Put $q = v - hau$. The nonexpansive velocity solve and (71) give

$$|w|, |q| \le c_w\|z\|, \qquad \|Tz\|, \|Az\| \le c_z\|z\|.$$

Since $|g(u) - au| \le K_3 u^2/2$, for $\|z\| \le 1$ we have

$$|w - q| \le hQ_0\|z\|^2, \qquad \|Tz - Az\| \le hC_N\|z\|^2. \tag{78}$$

The term $|w|^{5/2}$ is bounded by $c_w^{5/2}\|z\|^2$ only in this local estimate.

Let $r = \|z\| \le \rho$. An induction proves, through $j = N_h + 1$,

$$\|T^j z - A^j z\| \le 4B_0^3 T_0 C_N r^2 \le \varepsilon_0 r/2, \qquad \|T^j z\| \le 2B_0 r. \tag{79}$$

Indeed, earlier states in the latter ball have norm at most $2B_0\rho \le 1/2$, so (78) applies. Variation of constants bounds the accumulated defect by

$$B_0 \sum_{i=0}^{j-1} hC_N(2B_0 r)^2 \le 4B_0^3 T_0 C_N r^2.$$

The radius in (64) gives the first estimate, and $(B_0 + \varepsilon_0/2)r \le 2B_0 r$ closes the induction. This proves (75) without assuming capture in advance.

We next observe the sum of consecutive velocities over this window. Put

$$\omega = \sqrt{a}\sqrt{1 - ah^2/4}, \qquad X = \omega u, \qquad Y = v - hau/2.$$

Then $X^2 + Y^2 = 2Q(z)$, and A is the rotation

$$\begin{pmatrix} X' \\ Y' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}, \quad \cos\theta = 1 - ah^2/2, \quad \sin\theta = h\omega, \quad 0 < \theta \le \pi/3.$$

If $(\bar{u}_j, \bar{v}_j) = A^j z$, then $\bar{v}_j + \bar{v}_{j+1} = 2Y_j$. For a sinusoid of squared amplitude $2Q(z)$, the finite geometric sum gives

$$\sum_{j=0}^{N_h - 1} Y_j^2 \ge Q(z)\left(N_h - \frac{1}{\sin\theta}\right).$$

This also holds for the zero phase. Since $1/\omega \le 2/\sqrt{\mu} = \tau/2$ and $N_h h \ge \tau$, we obtain

$$h \sum_{j=0}^{N_h - 1} (\bar{v}_j + \bar{v}_{j+1})^2 \ge 2\tau Q(z) \ge \frac{m_0 \tau}{2} r^2 \ge c_0 r^2. \tag{80}$$

No initial angle, including a turning point, is excluded. From (79), the difference between an actual and linear observed sum is at most $\varepsilon_0 r$. The weighted Euclidean triangle inequality and the definition of ε_0 imply

$$h \sum_{j=0}^{N_h - 1} |v_j + v_{j+1}|^2 \ge (c_0/4) r^2.$$

Weighted Hölder, with total weight $N_h h \le T_0$, therefore gives

$$h \sum_{j=0}^{N_h - 1} |v_j + v_{j+1}|^p \ge T_0^{1-p/2} (c_0/4)^{p/2} r^p. \tag{81}$$

It remains to identify the actual energy loss. For $D = \psi'$, scalar uniform convexity gives

$$B_\psi(s,t) := \psi(s) - \psi(t) - D(t)(s-t) \geq \frac{2^{2-p}}{p}|s-t|^p. \tag{82}$$

For completeness, among intervals of length d, the integral of $(p-1)|t|^{p-2}$ is minimized by the interval centered at zero. Hence $(D(s) - D(t))(s-t) \geq 2^{2-p}|s-t|^p$. Integration along the segment from t to s proves (82); equal endpoints give zero without division.

The exact balance, obtained from (60), is

$$H(Tz) - H(z) = R_F(u,w) - \frac{h}{2}B_\psi(v,-w), \tag{83}$$

where

$$\begin{aligned} R_F(u,w) &= F(u+hw) - F(u) - \frac{hw}{2}[g(u) + g(u+hw)] \\ &= -\frac{h^3w^3}{2}\int_0^1 t(1-t)F'''(u+thw)\,dt. \end{aligned}$$

The signed integral identity follows by two integrations by parts and includes $w \leq 0$. Thus, for $\|z\| \leq 1$,

$$\left| R_F(u,w) + \frac{bh^3w^3}{12} \right| \leq \frac{K_4 c_w^3(1+c_w)}{12}h^3\|z\|^4. \tag{84}$$

The cubic correction in (69) has an exact fixed-step identity,

$$P(Az) - P(z) = \frac{bh^3}{12}(v - hau)^3. \tag{85}$$

To verify it, put $q = v - hau$ and $R(u,v) = u(2au^2 - 3ahuv + 3v^2)$. Substituting $v = q + ahu$ yields $R(u,v) = 2au^3 + 3ahu^2q + 3uq^2$. Expanding $R(u+hq, q)$ adds exactly $h(3-ah^2)q^3$, proving (85). This calculation does not invoke a normal-form theorem for the only C^2 friction map.

The sum of the absolute coefficients of P/h^2 is at most C_P. Consequently

$$|P(z)| \leq C_P h^2\|z\|^3, \qquad \|\nabla P(z)\| \leq 3C_P h^2\|z\|^2.$$

Both Az and Tz lie in the ball of radius $c_z\|z\|$. The mean-value estimate, (78), and $|w|, |q| \leq c_w\|z\|$ give

$$\begin{aligned} |P(Tz) - P(Az)| &\leq 6C_P c_z^2 Q_0 h^3\|z\|^4, \\ \frac{|b|h^3}{12}|q^3 - w^3| &\leq \frac{K_3 c_w^2 Q_0}{4}h^4\|z\|^4. \end{aligned}$$

Together with (82)–(85) and $h \leq 1$, these prove

$$J(Tz) - J(z) \leq -\frac{2^{2-p}}{2p}h|v+w|^p + C_R h^3\|z\|^4 \quad (\|z\| \leq 1). \tag{86}$$

The printed C_R is larger than the coefficient just required.

Sum (86) along the actual prefix controlled by (75), and use (81). This gives

$$J(T^{N_h}z) - J(z) \le -d_0 r^p + C_R h^2 T_0 (2B_0)^4 r^4 \le -\frac{d_0}{2} r^p.$$

The last inequality uses (66) and $r^{4-p} = \sqrt{r} \le \sqrt{\rho}$; the zero phase is immediate. This proves (76).

Finally, global smoothness and $F \ge 0$ give $g(u)^2 \le 2LF(u)$ by evaluating the descent inequality at $u - g(u)/L$. Hence

$$\frac{h}{2}|g(u)v| \le \frac{h\sqrt{L}}{2} E(u,v) \le \frac{1}{2} E(u,v), \qquad H(z) \ge \frac{m_0}{4}\|z\|^2.$$

On $\|z\| \le 2B_0\rho$, the radius restriction gives $|P(z)| \le (m_0/16)\|z\|^2$, so $J(z) \ge (m_0/8)\|z\|^2$. All terms in (70) are therefore nonnegative for $\|z\| \le \rho$, and its first term gives the lower bound in (77). The exact identity $\mathcal{W}_h(z) - \mathcal{W}_h(Tz) = J(z) - J(T^{N_h}z)$ gives its decrement. ∎

7.4 The global strict function and the actual limit

Define a continuous scalar cutoff on $[0, \infty)$ by

$$\chi(t) = \begin{cases} 1, & t \le b_1, \\ (b_2 - t)/(b_2 - b_1), & b_1 < t < b_2, \\ 0, & t \ge b_2, \end{cases}$$

and apply it to the actual losses:

$$\begin{aligned} \ell_h(z) = \chi(E(z))[\mathcal{W}_h(z) - \mathcal{W}_h(T_h z)] \\ + (1 - \chi(E(z)))[E(z) - E(T_h z)]. \end{aligned} \tag{87}$$

This is a blend of two signed decrements, not a cutoff of the raw cubic energy. Since $E(z) \le b_2$ implies $\|z\| \le \rho$, Lemmas 7 and 8 show that both losses are strictly positive on $b_1 < E < b_2$. On $E \le b_1$ the first loss is positive away from zero; on $E \ge b_2$ the second is positive. These statements include both cutoff endpoints. Thus ℓ_h is continuous, $\ell_h(0) = 0$ and $\ell_h(z) > 0$ for $z \ne 0$.

Let the first entrance index be

$$k_h(z) = \min\{k \ge 0 : E(T_h^k z) \le b_1\}.$$

Its finiteness, and compact-uniform boundedness, were already proved in Lemma 7. Set

$$\mathcal{V}_h(z) = \sum_{j=0}^{k_h(z)-1} \ell_h(T_h^j z) + \mathcal{W}_h(T_h^{k_h(z)} z). \tag{88}$$

An empty sum is zero. Before entrance, $k_h(T_h z) = k_h(z) - 1$. After entrance, the sublevel is invariant and $\ell_h = \mathcal{W}_h - \mathcal{W}_h \circ T_h$. Both cases give the exact identity (62). Every summand and terminal value in (88) is nonnegative. For a nonzero already-entered phase the terminal value is positive; otherwise the first loss is positive. Therefore $\mathcal{V}_h$ is positive definite.

We verify continuity without assuming continuity of the integer-valued entrance index. Fix a compact neighborhood K of any phase. By Lemma 7, a common finite integer M places every trajectory from K in $\{E \le b_1\}$ by step M. After first entrance the added losses telescope, so throughout K,

$$\mathcal{V}_h(z) = \sum_{j=0}^{M-1} \ell_h(T_h^j z) + \mathcal{W}_h(T_h^M z). \tag{89}$$

The right side is a fixed finite sum of continuous functions. This proves continuity even where the first entrance index changes. No common finite M over all initial phases is asserted.

If $E(z) > b_2$, let j_2 be the first entrance index into $\{E \le b_2\}$. It is finite. Before this index, $\chi = 0$, so the losses telescope in E. The remaining value of $\mathcal{V}_h$ is nonnegative; hence

$$\mathcal{V}_h(z) \ge E(z) - E(T_h^{j_2} z) \ge E(z) - b_2. \tag{90}$$

By (71), this proves global coercivity.

To finish the trajectory proof, let $k = k_h(z_0)$. For every $n \ge k$, the state remains in $\{E \le b_1\}$, and Lemma 8 applies. Its nonnegative local energy telescopes to give

$$\frac{d_0}{2} \sum_{n=k}^{\infty} \|z_n\|^{7/2} \le \mathcal{W}_h(z_k) < \infty.$$

The finite prefix adds a finite amount. This proves (61). For stability, continuity and $\mathcal{W}_h(0) = 0$ allow an initial ball so small that $E(z_0) \le b_1$ and $\mathcal{W}_h(z_0) < m_0 \varepsilon^2/8$. Invariance and (77) then imply

$$\|z_n\|^2 \le \frac{8}{m_0} \mathcal{W}_h(z_n) \le \frac{8}{m_0} \mathcal{W}_h(z_0) < \varepsilon^2 \quad (n \ge 0).$$

Thus the origin is stable as well as globally attractive, completing the proof of Theorem 3.

The cubic identity is exact in h, rather than a truncated small-step identity. The bound $ah^2 \le 1$ keeps its denominator away from zero and makes the finite observation window uniform. The remainder uses only four derivatives of the potential; the friction and its implicit map need only be C^2. The result is distinct from the earlier $0 < \alpha < 1$ theorems. It supplies neither their finite-length conclusion nor an arbitrary-exponent theorem. The structural constants are sufficient, not maximal or intended as practical tuning rules.

The function in (88) depends on a finite future entry segment, followed by the fixed window in (70). Its continuity proof is part of the result, but this formula is not presented as a new converse principle. General converse and finite-step machinery are credited above. The exact scalar trajectory assertion remains subject to the close-source limits described in Section 9; the unavailable MCRF body has not been excluded as a possible covering source.

8 Physical capture for nonconvex quartic potentials

The scalar mixed-drag estimate in Section 6 has a class-level counterpart. The step below depends on the polynomial coefficients and works for every original phase. It permits nonconvex potentials with unbounded negative curvature. Its conclusion is capture and approach to the critical set; it does not select one critical point.

Let $d \geq 1$ be finite and let

$$F(q) = \sum_{j=0}^{4} P_j(q), \qquad q \in \mathbb{R}^d, \tag{91}$$

where P_j is a real homogeneous polynomial of degree j and P_0 is constant. Suppose that, for every unit vector e,

$$\begin{aligned} P_4(e) \geq m > 0, \qquad |P_j(e)| \leq M_j \quad (j = 1, 2, 3), \\ \|\nabla^2 P_j(e)\| \leq H_j \quad (j = 2, 3, 4). \end{aligned} \tag{92}$$

The finite M_j, H_j are nonnegative upper bounds; they need not be exact extrema. Neither F nor P_4 is assumed convex. All norms in this section are Euclidean or their induced operator norms. Define

$$\begin{aligned} R &= \max\left\{1, \frac{3M_3}{m}, \sqrt{\frac{2M_2}{m}}, \left(\frac{M_1}{m}\right)^{1/3}\right\}, \\ L_q &= 1 + H_2 + H_3 + H_4, \\ K_F &= \frac{3}{4} M_1^{4/3} m^{-1/3} + \frac{M_2^2}{m} + \frac{27 M_3^4}{4m^3}, \\ C_\eta &= R^2 + 2m^{-2/3} + \frac{1}{m\sqrt{\eta}} \quad (\eta > 0), \\ C_q &= C_{1/(4L_q)}, \qquad h_q = \min\left\{1, \frac{1}{2L_q(C_q + 1)}\right\}. \end{aligned} \tag{93}$$

These constants are chosen before the initial phase. The literal map is

$$(1 + h)w + h\|w\|^2 w = v - h\nabla F(q), \qquad T_h(q, v) = (q + hw, w). \tag{94}$$

The force is at the old position and the mixed radial drag at the new velocity. Set

$$A(q, w) = \nabla F(q) + w + \|w\|^2 w, \qquad \mathcal{E}(q, v) = F(q) - P_0 + K_F + \frac{1}{2}\|v\|^2. \tag{95}$$

The shift in $\mathcal{E}$ requires no unknown minimum value.

Lemma 9. *Under* (92), *F is coercive and its critical set is nonempty, compact and contained in $\{\|q\| < R\}$. For every q, v,*

$$\begin{aligned} F(q) &\geq P_0 + \frac{m}{4}\|q\|^4 - K_F, \qquad \mathcal{E}(q,v) \geq \frac{m}{4}\|q\|^4 + \frac{1}{2}\|v\|^2, \\ \|\nabla^2 F(q)\| &\leq L_q(1+\|q\|^2), \qquad \|\nabla F(q)\| \geq m\|q\|^3 \quad \textit{if } \|q\| \geq R. \end{aligned} \tag{96}$$

For every $\eta > 0$ and every q, w one also has

$$\|q\|^2\|w\|^2 \leq \eta\|A(q,w)\|^2 + C_\eta(\|w\|^2 + \|w\|^4). \tag{97}$$

Proof. Write $r = \|q\|$. Maximizing $M_j r^j - mr^4/4$ over $r \geq 0$ gives, respectively,

$$\frac{3}{4}M_1^{4/3}m^{-1/3}, \qquad \frac{M_2^2}{m}, \qquad \frac{27M_3^4}{4m^3}.$$

The maximizing radii are $(M_1/m)^{1/3}$, $(2M_2/m)^{1/2}$ and $3M_3/m$; zero coefficients give zero maxima. Subtracting the three resulting bounds from mr^4 proves the first two inequalities in (96). In particular, F attains a global minimum and has a critical point. Homogeneity gives

$$\|\nabla^2 F(q)\| \leq H_2 + H_3 r + H_4 r^2 \leq L_q(1+r^2).$$

This includes $q = 0$, where the cubic and quartic Hessians vanish. Euler's identity gives, when $r \geq R$,

$$\langle q, \nabla F(q)\rangle \geq 4mr^4 - 3M_3 r^3 - 2M_2 r^2 - M_1 r \geq mr^4.$$

Each subtracted term is at most mr^4 by the definition of R. Since $R \geq 1$, division by r proves the gradient bound. The closed critical set is therefore bounded and lies strictly inside that sphere.

To prove (97), put $a = \|A(q,w)\|$ and $b = \|w\|$. The radial estimate, the triangle inequality and subadditivity of $t^{2/3}$ imply

$$r^2 b^2 \leq R^2 b^2 + m^{-2/3}\big(a^{2/3}b^2 + b^{8/3} + b^4\big).$$

For $r < R$ the first term suffices; for $r \geq R$ use $mr^3 \leq \|\nabla F(q)\| \leq a + b + b^3$. Young's inequality yields

$$m^{-2/3}a^{2/3}b^2 \leq \frac{\eta}{3}a^2 + \frac{2}{3m\sqrt{\eta}}b^3 \leq \eta a^2 + \frac{1}{m\sqrt{\eta}}(b^2 + b^4).$$

Finally $b^{8/3} \leq b^2 + b^4$ for all $b \geq 0$. Combining these estimates proves (97) with the stated conservative common coefficient. No velocity or force norm is divided out. ∎

Theorem 4. *Assume* (91)–(92). *For every $h > 0$, the update* (94) *defines a unique continuous global map. For every fixed $0 < h \leq h_q$ and every original phase (q, v), its actual next velocity satisfies*

$$\mathcal{E}(T_h(q,v)) \leq \mathcal{E}(q,v) - \frac{h}{2}(\|w\|^2 + \|w\|^4) - \frac{1}{4}\|v - w\|^2. \tag{98}$$

The energy is coercive on the full phase space and decreases strictly away from $\operatorname{crit} F \times \{0\}$. *For arbitrary independent* q_0, v_0, *let* $(q_n, v_n) = T_h^n(q_0, v_0)$ *and* $\mathcal{E}_0 = \mathcal{E}(q_0, v_0)$. *Then, for every* $n \geq 0$,

$$0 \leq \mathcal{E}(q_n, v_n) \leq \mathcal{E}_0, \qquad \|q_n\|^4 \leq \frac{4\mathcal{E}_0}{m}, \qquad \|v_n\|^2 \leq 2\mathcal{E}_0. \tag{99}$$

Moreover,

$$\begin{aligned} \sum_{n=0}^{\infty} (\|v_{n+1}\|^2 + \|v_{n+1}\|^4) &\leq \frac{2\mathcal{E}_0}{h}, \\ \sum_{n=0}^{\infty} \|v_{n+1} - v_n\|^2 &\leq 4\mathcal{E}_0. \end{aligned} \tag{100}$$

Consequently $v_n \to 0$, $\nabla F(q_n) \to 0$ *and*

$$\operatorname{dist}\big((q_n, v_n), \operatorname{crit} F \times \{0\}\big) \longrightarrow 0. \tag{101}$$

Proof. For any right side $z = v - h\nabla F(q)$, the new velocity is the unique minimizer of

$$\frac{1+h}{2}\|w\|^2 + \frac{h}{4}\|w\|^4 - \langle z, w\rangle.$$

Its Hessian is $(1 + h + h\|w\|^2)I + 2hww^{\mathsf{T}} \succeq (1+h)I$. The functional is coercive, and the inverse is continuous by strong monotonicity. Thus the step is uniquely defined at every phase, for every $h > 0$. A fixed phase necessarily has $w = v = 0$ and $\nabla F(q) = 0$, and every such phase is fixed.

We prove the decrement before using boundedness. Put $r = \|q\|$, $b = \|w\|$ and $a = \|A(q, w)\|$. The exact original velocity relation is $v - w = hA(q, w)$. Taylor's integral remainder and (96) give

$$F(q + hw) - F(q) - h\langle \nabla F(q), w\rangle \leq \frac{L_q h^2}{2}(1 + 2r^2 + 2h^2b^2)b^2,$$

since $\|q + thw\|^2 \leq 2r^2 + 2h^2b^2$ for $0 \leq t \leq 1$. The kinetic identity is

$$\frac{1}{2}(\|w\|^2 - \|v\|^2) = -h\langle \nabla F(q), w\rangle - h(b^2 + b^4) - \frac{h^2}{2}a^2.$$

After cancellation, apply (97) with $\eta = 1/(4L_q)$ to obtain

$$\begin{aligned} \mathcal{E}(T_h(q, v)) - \mathcal{E}(q, v) \leq\ & -h(b^2 + b^4) - \frac{h^2}{4}a^2 \\ & + L_q h^2\big[(C_q + 1/2)b^2 + (C_q + h^2)b^4\big]. \end{aligned} \tag{102}$$

Now $h \leq 1$ and $L_q h(C_q + 1) \leq 1/2$ prove (98), including the upper endpoint. Equality of successive energy values forces $w = 0$ and $v = w$, hence $\nabla F(q) = 0$. Conversely a fixed phase gives equality. In particular, the estimate also treats a turning step $w = 0$; its exact energy drop is $\|v\|^2/2$.

Only now iterate the inequality. The coercive lower bound in (96) proves (99); telescoping and nonnegativity prove both sums in (100). Hence $v_n \to 0$ and $v_n - v_{n+1} \to 0$. The unchanged map gives

$$\nabla F(q_n) = \frac{v_n - v_{n+1}}{h} - v_{n+1} - \|v_{n+1}\|^2 v_{n+1} \longrightarrow 0.$$

Every cluster point of the bounded phase orbit therefore belongs to $\operatorname{crit} F \times \{0\}$. If (101) failed, a subsequence separated from that closed set would have a convergent further subsequence in it, a contradiction. ∎

The next statement gives one compact absorbing set for all initial phases. Its entry time, as expected, depends on the initial energy.

Corollary 8.1. *Under the hypotheses of Theorem 4, define*

$$\begin{aligned} S_R &= M_1 R + M_2 R^2 + M_3 R^3, \\ E_{\mathrm{cap}} &= K_F + \frac{H_4}{12} R^4 + S_R + \frac{1}{2}, \\ b_0 &= \min\{1/2, mR^3/4\}, \\ \delta_h &= \min\{hb_0^2/2,\ h^2 m^2 R^6/16,\ 1/16\} > 0. \end{aligned} \tag{103}$$

For every $0 < h \le h_q$, an original phase with $\mathcal{E}(q,v) > E_{\mathrm{cap}}$ satisfies

$$\mathcal{E}(T_h(q,v)) \le \mathcal{E}(q,v) - \delta_h. \tag{104}$$

The compact set $\{\mathcal{E} \le E_{\mathrm{cap}}\}$ is forward invariant and absorbs every original orbit. More precisely, for

$$N = \left\lceil \frac{\max\{\mathcal{E}_0 - E_{\mathrm{cap}}, 0\}}{\delta_h} \right\rceil, \tag{105}$$

one has $\mathcal{E}(q_n, v_n) \le E_{\mathrm{cap}}$ for all $n \ge N$. The entry bound is uniform over each bounded set of initial phases for a fixed h.

Proof. Twice differentiating $P_4(te) = t^4 P_4(e)$ at $t = 1$ gives $12P_4(e) = e^{\mathsf{T}} \nabla^2 P_4(e) e \le H_4$. Thus if $\|q\| \le R$ and $\|v\| \le 1$, homogeneity gives $\mathcal{E}(q,v) \le E_{\mathrm{cap}}$. Above the cap, therefore, either $\|q\| > R$ or $\|v\| > 1$.

If $\|w\| \ge b_0$, the first loss in (98) is at least $hb_0^2/2$. Otherwise $\|w\| < b_0 \le 1/2$ and

$$\|w + \|w\|^2 w\| \le 2\|w\| \le 2b_0 \le mR^3/2.$$

If $\|q\| > R$, the force bound gives $\|A(q,w)\| \ge mR^3/2$; since $v - w = hA(q,w)$, the jump loss is at least $h^2 m^2 R^6/16$. In the remaining case $\|v\| > 1$, one has $\|v - w\| \ge 1/2$, giving loss at least $1/16$. This proves (104) directly, without using the asymptotic conclusions of the theorem.

Coercivity and descent make the sublevel compact and invariant. If $\mathcal{E}_0 \le E_{\mathrm{cap}}$, then $N = 0$. Otherwise either entry occurs before N, or every index $0, \dots, N-1$ lies above the cap. In the latter case, summing (104) exactly N times gives

$$\mathcal{E}(q_N, v_N) \le \mathcal{E}_0 - N\delta_h \le E_{\mathrm{cap}}.$$

Equality is entry, so an integer quotient needs no additional step. Invariance proves the assertion at every later index. Finally, the continuous energy is bounded on the closure of any bounded initial set, which proves the stated uniformity. Neither uniform entry over unbounded initial phases nor uniformity as $h \downarrow 0$ is asserted. ∎

Example 1. In dimension two let

$$F(x,y) = x^4 + y^4 - x^2y^2 - \frac{x^2+y^2}{2}.$$

For $r^2 = x^2 + y^2$, the exact identities

$$P_4 - \frac{1}{4}r^4 = \frac{3}{4}(x^2-y^2)^2, \qquad F + \frac{1}{4} = \frac{1}{4}(r^2-1)^2 + \frac{3}{4}(x^2-y^2)^2$$

give a positive leading part and four global minima $x^2 = y^2 = 1/2$, of value $-1/4$. At $(t,0)$ the transverse Hessian eigenvalue is $-2t^2 - 1$, so the negative curvature is unbounded. Valid data are

$$m = 1/4, \quad M_2 = 1/2, \quad H_2 = 1, \quad H_4 = 14, \qquad M_1 = M_3 = H_3 = 0.$$

Indeed the quartic Hessian on the unit circle has eigenvalues $5 \pm \sqrt{49 - 180x^2y^2}$, so its exact norm supremum is 12 and 14 is a conservative bound. The definitions give

$$R = 2, \quad L_q = 16, \quad K_F = 1, \quad C_q = 36 + 2\,4^{2/3} < 42.$$

Thus every $0 < h \le 1/1376$ is admissible. The energy is $F + 1 + \|v\|^2/2$ and the common cap is $133/6$. The example makes no further assertion about selection among its critical points.

The physical energy and its velocity-jump identity are classical. The scalar two-region argument behind Proposition 6.1 is direct ancestry: drag controls the growing curvature when the new velocity is large, and the residual square controls the complementary region. Lemma 9 gives that absorption for a positive-leading quartic class. Its conservative step does not improve the sharper step of the fixed scalar proposition.

The exact map is contained in [2, Section 8]; its bounded-orbit and attained-ball step premises do not supply the present structural all-phase bound. Even after capture, its point-convergence theorem cannot be applied without checking the additional step premise for the same fixed h. The uniform-step stencil of [3] also coincides after setting

$(q, v) = (u^n, v^{n-1})$. Its nonlinear restoring perturbation has growth at most $2(p-1)/p$, which is $3/2$ for cubic drag ($p = 4$), and thus excludes the present cubic restoring force. Its estimates concern a fixed final time and mesh refinement. The comparison uses the specified full author version. Different fully implicit and polarized schemes have their own energy and convergence results; these do not change the force evaluation in (94). The material source-access limits stated below remain. No point convergence, finite length, stability of the entire critical set, maximal step or exhaustive priority is claimed for this class.

9 Physical energy and the scope of the conclusion

The distinction between E and Φ is already visible for a quadratic potential. Take $d = 2$, $F(u) = \|u\|^2/2$, $\alpha = 1/2$ and $h = L = 1$. For $t > 0$, set

$$u_0 = 0, \qquad v_0 = (t^2 + t^3, 0), \qquad v_1 = (t^2, 0), \qquad u_1 = v_1.$$

These are exact original states of (4), since $g(u_0) = 0$ and $\|v_1\|^{1/2} v_1 = (t^3, 0)$. Direct calculation gives

$$E(u_1, v_1) - E(u_0, v_0) = \frac{t^4}{2} - t^5 - \frac{t^6}{2} > 0 \quad (0 < t < \sqrt{2} - 1). \tag{106}$$

The initial energy tends to zero with t. Hence physical-energy monotonicity cannot replace the Lyapunov estimate, even in an arbitrarily small phase neighborhood. This calculation concerns the energy sign; it does not contradict the trajectory convergence in Theorem 1.

For a nonisolated minimum set, consider the source potential

$$F(u) = \big(\|u - (1,1)\|^2 - 1\big)^2 \quad \text{on } \mathbb{R}^2.$$

Its full minimum set is the unit circle about $(1, 1)$, with normal Hessian eigenvalue 8 and tangent kernel. On the ball of radius 3 about $(1, 1)$, its Hessian norm is at most 104. Thus $\alpha = 1/2$ and $h = 1/100$, the parameters in [1, author version, equation (55)], satisfy (3). Theorem 1 provides a full sufficiently small-energy basin for this potential and these parameters. The cited numerical example does not specify its initial coordinates; no identification of that plotted orbit with this basin is made. Its quartic potential has an unbounded Hessian, so it is outside the global class of Theorem 2.

In Theorem 1, the range $0 < \alpha < 1$ has two explicit uses: the signed force remainder in (26) has cubic order, and the velocity rate in (9) is summable. Neither argument supplies the endpoint $\alpha = 1$. That theorem is local in the original phase energy and does not select a particular point on the minimum manifold. Theorem 2 provides arbitrary-start capture under its additional global curvature and step assumptions; it does not transfer the local finite-length or point-convergence conclusion to those starts.

Earlier discrete-angle arguments remain important ancestry. Alaa and Pierre [11, Section 4, Theorem 4.5] treat a linearly implicit gradient-like scheme with linear velocity damping and quantitative descent assumptions. Their result does not permit replacement of that damping term by the present pure power law. The comparison here uses the indexed text of their final article; a local PDF render was not available. Horsin and Jendoubi's abstract powered-descent theorem [1, Theorem 2.1] is another predecessor for the finite-length conclusion once a suitable estimate is established. The proof of Theorem 1 derives that estimate and the local capture of (4) directly.

The source comparison uses the specified HJ author version and the inspected final ABC, symplectic-energy and finite-step sources. The detailed final-journal HJ body and the full text of Cheikh's forced-discretization article [14] were not available for comparison. The MCRF publisher's complete-text route remains restricted, so a covering result in that unseen body has not been excluded. The pure-power results comprise the local trajectory theorem, the separately hypothesized global energy theorem and the scalar strong-convexity result in Section 7. The conservative commutator, finite-step construction and convergence machinery are attributed above. Proposition 6.1 gives a separate physical-energy estimate for one mixed-drag scalar model. Section 8 adds a positive-leading quartic class under the mixed radial drag, with a structural step and critical-set approach only. No exhaustive priority claim is made.

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